

SI 140A-02 Probability & Statistics for EECS, Fall 2025

Homework 4

Name:

Student ID:

Due on Oct.20, 2025, 23:59 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

Consider the Monty Hall problem, except that Monty enjoys opening door 2 more than he enjoys opening door 3, and if he has a choice between opening these two doors, he opens door 2 with probability p , where $\frac{1}{2} \leq p \leq 1$.

To recap: there are three doors, behind one of which there is a car (which you want), and behind the other two of which there are goats (which you don't want). Initially, all possibilities are equally likely for where the car is. You choose a door, which for concreteness we assume is door 1. Monty Hall then opens a door to reveal a goat, and offers you the option of switching. Assume that Monty Hall knows which door has the car, will always open a goat door and offer the option of switching, and as above assume that if Monty Hall has a choice between opening door 2 and door 3, he chooses door 2 with probability p (with $\frac{1}{2} \leq p \leq 1$).

- (a) Find the unconditional probability that the strategy of always switching succeeds (unconditional in the sense that we do not condition on which of doors 2 or 3 Monty opens).
- (b) Find the probability that the strategy of always switching succeeds, given that Monty opens door 2.
- (c) Find the probability that the strategy of always switching succeeds, given that Monty opens door 3.

Problem 2

A sequence of $n \geq 1$ independent trials is performed, where each trial ends in “success” or “failure” (but not both). Let p_i be the probability of success in the i th trial, $q_i = 1 - p_i$, and $b_i = q_i - \frac{1}{2}$, for $i = 1, 2, \dots, n$. Let A_n be the event that the number of successful trials is even.

(a) Show that for $n = 2$, $P(A_2) = \frac{1}{2} + 2b_1b_2$.

(b) Show by induction that

$$P(A_n) = \frac{1}{2} + 2^{n-1}b_1b_2 \cdots b_n.$$

(This result is useful in cryptography. Also, note that it implies that if n coins are flipped, then the probability of an even number of Heads is $1/2$ if and only if at least one of the coins is fair.) *Hint:* Group some trials into a supertrial.

(c) Check directly that the result of (b) is true in the following simple cases: $p_i = \frac{1}{2}$ for some i ; $p_i = 0$ for all i ; $p_i = 1$ for all i .

Problem 3

Nick and Penny are independently performing independent Bernoulli trials. For concreteness, assume that Nick is flipping a nickel with probability p_1 of Heads and Penny is flipping a penny with probability p_2 of Heads. Let $X_1, X_2, \dots$ be Nick's results and $Y_1, Y_2, \dots$ be Penny's results, with $X_i \sim \text{Bern}(p_1)$ and $Y_j \sim \text{Bern}(p_2)$.

(a) Find the distribution and expected value of the first time at which they are simultaneously successful, i.e., the smallest n such that $X_n = Y_n = 1$.

Hint: Define a new sequence of Bernoulli trials and use the story of the Geometric.

(b) Find the expected time until at least one has a success (including the success).

Hint: Define a new sequence of Bernoulli trials and use the story of the Geometric.

(c) For $p_1 = p_2$, find the probability that their first successes are simultaneous, and use this to find the probability that Nick's first success precedes Penny's.

Problem 4

For X and Y binary digits (0 or 1), let $X \oplus Y$ be 0 if $X = Y$ and 1 if $X \neq Y$ (this operation is called exclusive or (often abbreviated to XOR), or addition mod 2).

- (a) Let $X \sim \text{Bern}(p)$ and $Y \sim \text{Bern}(1/2)$, independently. What is the distribution of $X \oplus Y$?
- (b) With notation as in sub-problem (a), is $X \oplus Y$ independent of X ? Is $X \oplus Y$ independent of Y ? Be sure to consider both the case $p = 1/2$ and the case $p \neq 1/2$.
- (c) Let $X_1, \dots, X_n$ be i.i.d. (i.e., independent and identically distributed) $\text{Bern}(1/2)$ R.V.s. For each nonempty subset J of $\{1, 2, \dots, n\}$, let

$$Y_J = \bigoplus_{j \in J} X_j.$$

Show that $Y_J \sim \text{Bern}(1/2)$ and that these $2^n - 1$ R.V.s are pairwise independent, but not independent.

Problem 5

Let X be a discrete R.V. whose distinct possible values are $a_1, a_2, \dots, a_n$, with probabilities $p_1, p_2, \dots, p_n$, respectively (so $p_1 + p_2 + \dots + p_n = 1$). The entropy of X is defined to be the average surprise of learning the value of X :

$$H(X) = \sum_{j=1}^n p_j \log_2(1/p_j).$$

Show that the maximum entropy for X is when its distribution is uniform over $a_1, a_2, \dots, a_n$, i.e., $p_j = 1/n, \forall j$.